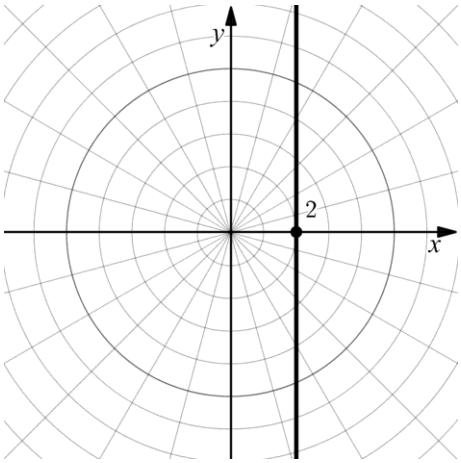
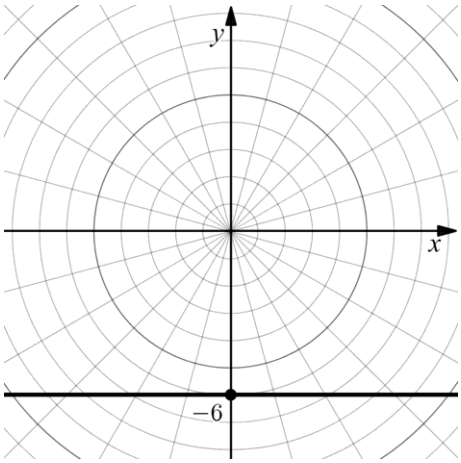
1) a)

$r = 2\sec\theta \Rightarrow r = \frac{2}{\cos\theta} \Rightarrow r\cos\theta = 2$



1) b)

$r = -6\,\text{cosec}\,\theta \Rightarrow r = \frac{-6}{\sin\theta} \Rightarrow r\sin\theta = -6$



2) a)

$r = 2\sec\left(\frac{\pi}{4} - \theta\right) \Rightarrow r\cos\left(\frac{\pi}{4} - \theta\right) = 2$

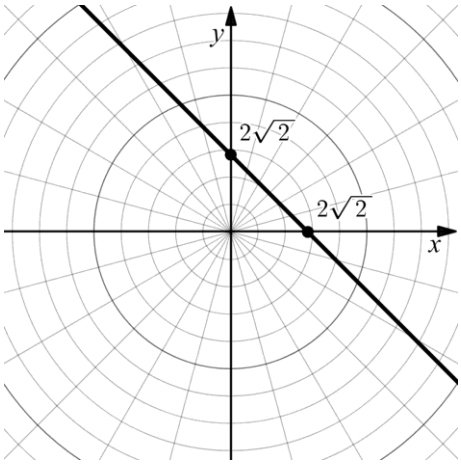
$r\cos\frac{\pi}{4}\cos\theta + r\sin\frac{\pi}{4}\sin\theta = 2$

$\frac{\sqrt{2}}{2}r\cos\theta + \frac{\sqrt{2}}{2}r\sin\theta = 2$

$\frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y = 2$

$\frac{\sqrt{2}}{2}y = -\frac{\sqrt{2}}{2}x + 2$

$y = -x + 2\sqrt{2}$



2) b)

$r = 2\operatorname{cosec}\left(\frac{\pi}{4} - \theta\right) \Rightarrow r\sin\left(\frac{\pi}{4} - \theta\right) = 2$

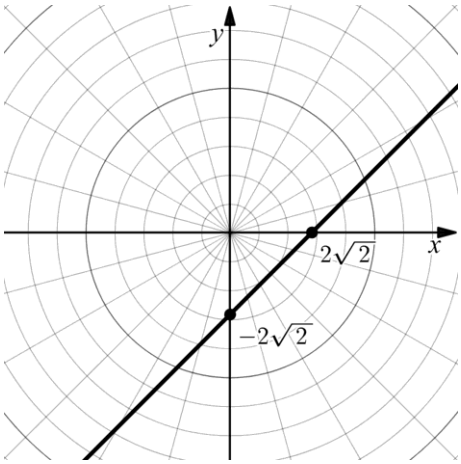
$r\sin\frac{\pi}{4}\cos\theta - r\cos\frac{\pi}{4}\sin\theta = 2$

$\frac{\sqrt{2}}{2}r\cos\theta - \frac{\sqrt{2}}{2}r\sin\theta = 2$

$\frac{\sqrt{2}}{2}x - \frac{\sqrt{2}}{2}y = 2$

$\frac{\sqrt{2}}{2}y = \frac{\sqrt{2}}{2}x - 2$

$y = x - 2\sqrt{2}$



2) c)

$r = -3\operatorname{cosec}\left(\frac{\pi}{6} - \theta\right) \Rightarrow r\sin\left(\frac{\pi}{6} - \theta\right) = -3$

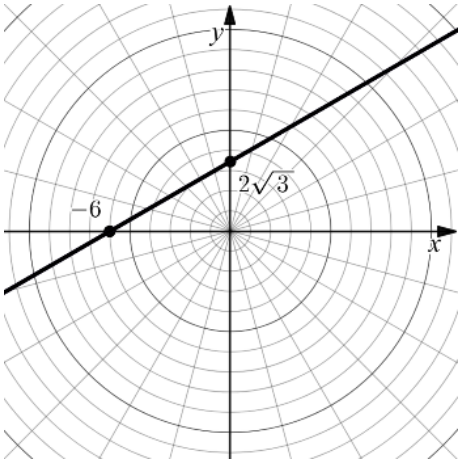
$r\sin\frac{\pi}{6}\cos\theta - r\cos\frac{\pi}{6}\sin\theta = -3$

$\frac{1}{2}r\cos\theta - \frac{\sqrt{3}}{2}r\sin\theta = -3$

$\frac{1}{2}x - \frac{\sqrt{3}}{2}y = -3$

$\frac{\sqrt{3}}{2}y = \frac{1}{2}x + 3$

$y = \frac{\sqrt{3}}{3}x + 2\sqrt{3}$



2) d)

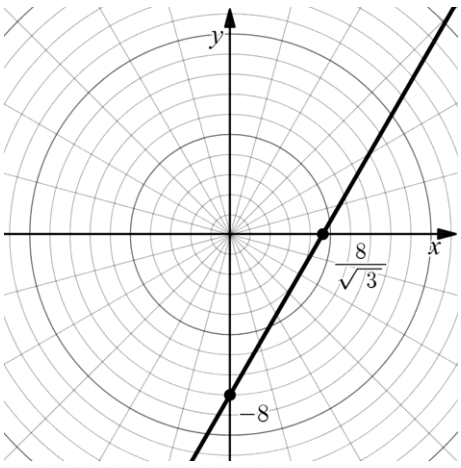
$r = -4\sec\left(\frac{5\pi}{6} - \theta\right) \Rightarrow r\cos\left(\frac{5\pi}{6} - \theta\right) = -4$

$r\cos\frac{5\pi}{6}\cos\theta + r\sin\frac{5\pi}{6}\sin\theta = -4$

$-\frac{\sqrt{3}}{2}r\cos\theta + \frac{1}{2}r\sin\theta = -4$

$-\frac{\sqrt{3}}{2}x + \frac{1}{2}y = -4$

$y = \sqrt{3}x - 8$



3) a)

$r = p\sec(a-\theta) \Rightarrow r\cos(a-\theta) = p \Rightarrow r\cos a\cos\theta + r\sin a\sin\theta = p$

$m = -\frac{6}{\frac{6}{\sqrt{3}}} = -\sqrt{3}$

$y = -\sqrt{3}x - 6$

$r\sin\theta = -\sqrt{3}r\cos\theta - 6$

$r(\sin\theta + \sqrt{3}\cos\theta) = -6$

Now find $\sin\theta + \sqrt{3}\cos\theta$:

$\sqrt{1^2 + (\sqrt{3})^2} = 2$ and $\theta = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

So $\sin\theta + \sqrt{3}\cos\theta = 2\cos\left(\theta - \frac{\pi}{6}\right)$

Replacing in $r(\sin\theta + \sqrt{3}\cos\theta) = -6$ gives $2r\cos\left(\theta - \frac{\pi}{6}\right) = -6$

Cosine is even so $2r\cos\left(\frac{\pi}{6} - \theta\right) = -6$

$\therefore r\cos\left(\frac{\pi}{6} - \theta\right) = -3$

And so since $r = p\sec(a-\theta) \Rightarrow r\cos(a-\theta) = p$

then $r\cos\left(\frac{\pi}{6} - \theta\right) = -3 \Rightarrow r = -3\sec\left(\frac{\pi}{6} - \theta\right)$

$\therefore a = \frac{\pi}{6}$ and $p = -3$

3) b)

$r = p\sec(a-\theta) \Rightarrow r\cos(a-\theta) = p \Rightarrow r\cos a\cos\theta + r\sin a\sin\theta = p$

$m = -\frac{\frac{8}{\sqrt{3}}}{8} = -\frac{\sqrt{3}}{3}$

$y = -\frac{\sqrt{3}}{3}x + \frac{8}{\sqrt{3}}$

$r\sin\theta = -\frac{\sqrt{3}}{3}r\cos\theta + \frac{8}{\sqrt{3}}$

$r\left(\sin\theta + \frac{\sqrt{3}}{3}\cos\theta\right) = \frac{8}{\sqrt{3}}$

Now find $\sin\theta + \frac{\sqrt{3}}{3}\cos\theta$:

$\sqrt{1^2 + \left(\frac{\sqrt{3}}{3}\right)^2} = \frac{2\sqrt{3}}{3}$ and $\theta = \arctan\left(\frac{3}{\sqrt{3}}\right) = \frac{\pi}{3}$

So $\sin\theta + \sqrt{3}\cos\theta = \frac{2\sqrt{3}}{3}\cos\left(\theta - \frac{\pi}{3}\right)$

Replacing in $r\left(\sin\theta + \sqrt{3}\cos\theta\right) = \frac{8}{\sqrt{3}}$ gives $\frac{2\sqrt{3}}{3}r\cos\left(\theta - \frac{\pi}{3}\right) = \frac{8}{\sqrt{3}} \Rightarrow 2r\cos\left(\theta - \frac{\pi}{3}\right) = 8$

Cosine is even so $2r\cos\left(\frac{\pi}{3} - \theta\right) = 8$

$\therefore r\cos\left(\frac{\pi}{3} - \theta\right) = 4$

And so since $r = p\sec(a-\theta) \Rightarrow r\cos(a-\theta) = p$

then $r\cos\left(\frac{\pi}{3} - \theta\right) = 4 \Rightarrow r = 4\sec\left(\frac{\pi}{3} - \theta\right)$

$\therefore a = \frac{\pi}{3}$ and $p = 4$

3) c)

$r = p\sec(a-\theta) \Rightarrow r\cos(a-\theta) = p \Rightarrow r\cos a\cos\theta + r\sin a\sin\theta = p$

$m = 1$

$y = x + 2\sqrt{2}$

$r\sin\theta = r\cos\theta + 2\sqrt{2}$

$r(\sin\theta - \cos\theta) = 2\sqrt{2}$

Now find $\sin\theta - \cos\theta$:

$\sqrt{1^2 + (-1)^2} = \sqrt{2}$ and $\theta = \arctan\left(-\frac{1}{1}\right) = -\frac{\pi}{4}$

But since it is the curve for $0 \leq \theta \leq 2\pi$ then use $\frac{3\pi}{4}$:

So $\sin\theta - \cos\theta = \sqrt{2}\cos\left(\theta - \frac{3\pi}{4}\right)$

Replacing in $r(\sin\theta - \cos\theta) = 2\sqrt{2}$ gives $\sqrt{2}r\cos\left(\theta - \frac{3\pi}{4}\right) = 2\sqrt{2}$

Cosine is even so $\sqrt{2}r\cos\left(\frac{3\pi}{4} - \theta\right) = 2\sqrt{2}$

$\therefore r\cos\left(\frac{3\pi}{4} - \theta\right) = 2$

And so since $r = p\sec(a-\theta) \Rightarrow r\cos(a-\theta) = p$

then $r\cos\left(\frac{3\pi}{4} - \theta\right) = 2 \Rightarrow r = 2\sec\left(\frac{3\pi}{4} - \theta\right)$

$\therefore a = \frac{3\pi}{4}$ and $p = 2$

4) a)

$r = p\,\text{cosec}(a-\theta) \Rightarrow r\sin(a-\theta) = p \Rightarrow r\sin a\cos\theta - r\cos a\sin\theta = p$

$m = \frac{\frac{4}{\sqrt{2}}}{\frac{4}{\sqrt{2}}} = 1$

$y = x - 4\sqrt{2}$

$r\sin\theta = r\cos\theta - 4\sqrt{2}$

$r(\sin\theta - \cos\theta) = -4\sqrt{2}$

Now find $\sin\theta - \cos\theta$:

$\sqrt{1^2+1^2} = \sqrt{2}$ and $\theta = \arctan\left(\frac{1}{1}\right) = \frac{\pi}{4}$

So $\sin\theta - \cos\theta = \sqrt{2}\sin\left(\theta - \frac{\pi}{4}\right)$

Replacing in $r(\sin\theta - \cos\theta) = -4\sqrt{2}$ gives $\sqrt{2}r\sin\left(\theta - \frac{\pi}{4}\right) = -4\sqrt{2}$

Sine is odd so $-r\sin\left(\frac{\pi}{4} - \theta\right) = -4$

$\therefore r\sin\left(\frac{\pi}{4} - \theta\right) = 4$

And so since $r = p\,\text{cosec}(a-\theta) \Rightarrow r\sin(a-\theta) = p$

then $r\sin\left(\frac{\pi}{4} - \theta\right) = 4 \Rightarrow r = 4\,\text{cosec}\left(\frac{\pi}{4} - \theta\right)$

$\therefore a = \frac{\pi}{4}$ and $p = 4$

4) b)

$r = p\,\text{cosec}(a-\theta) \Rightarrow r\sin(a-\theta) = p \Rightarrow r\sin a\cos\theta - r\cos a\sin\theta = p$

$m = -\frac{2\sqrt{3}}{6} = -\frac{\sqrt{3}}{3}$

$y = -\frac{\sqrt{3}}{3}x + 2\sqrt{3}$

$r\sin\theta = -\frac{\sqrt{3}}{3}r\cos\theta + 2\sqrt{3}$

$r\left(\sin\theta + \frac{\sqrt{3}}{3}\cos\theta\right) = 2\sqrt{3}$

Now find $\sin\theta + \frac{\sqrt{3}}{3}\cos\theta$:

$\sin(\theta - A) \equiv \sin\theta\cos A - \cos\theta\sin A$

$\sqrt{1^2 + \left(\frac{\sqrt{3}}{3}\right)^2} = \frac{2}{\sqrt{3}}$ and $\theta = \arctan\left(-\frac{\sqrt{3}}{3}\right) = -\frac{\pi}{6}$

So $\sin\theta + \frac{\sqrt{3}}{3}\cos\theta = \frac{2}{\sqrt{3}}\sin\left(\theta + \frac{\pi}{6}\right)$

Replacing in $r\left(\sin\theta + \frac{\sqrt{3}}{3}\cos\theta\right) = 2\sqrt{3}$ gives $\frac{2}{\sqrt{3}}r\sin\left(\theta + \frac{\pi}{6}\right) = 2\sqrt{3}$

But $\sin\left(\theta + \frac{\pi}{6}\right) = \sin\left(\frac{5\pi}{6} - \theta\right)$

So $\frac{2}{\sqrt{3}}r\sin\left(\frac{5\pi}{6} - \theta\right) = 2\sqrt{3}$

$\therefore r\sin\left(\frac{5\pi}{6} - \theta\right) = 3$

And so since $r = p\,\text{cosec}(a-\theta) \Rightarrow r\sin(a-\theta) = p$

then $r\sin\left(\frac{5\pi}{6} - \theta\right) = 3 \Rightarrow r = 3\,\text{cosec}\left(\frac{5\pi}{6} - \theta\right)$

$\therefore a = \frac{5\pi}{6}$ and $p = 3$

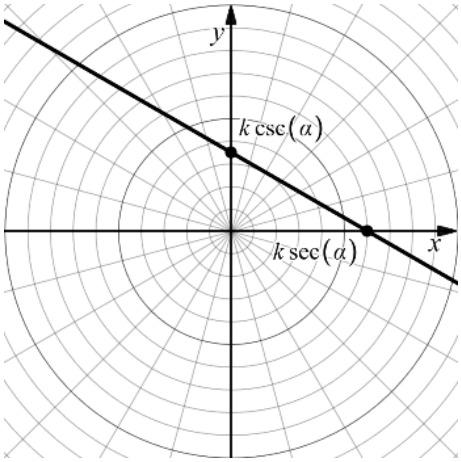
5) a)

$r\cos(\alpha-\theta)=k$

$r\cos\alpha\cos\theta+r\sin\alpha\sin\theta=k$

$x\cos\alpha+y\sin\alpha=k$

$y=-\frac{\cos\alpha}{\sin\alpha}x+\frac{k}{\sin\alpha}$

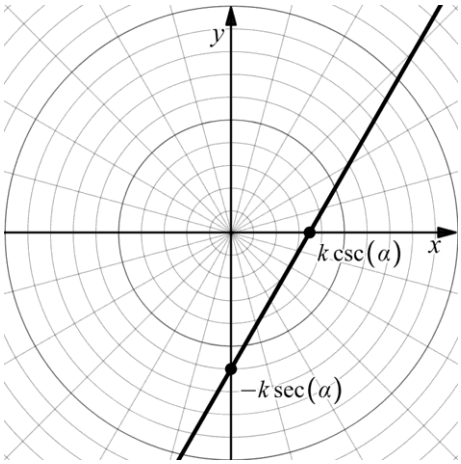


5) b)

$\sin(\alpha-\theta)=k$

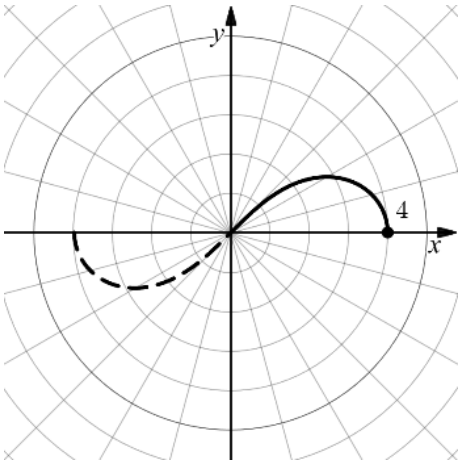
$r\sin\alpha\cos\theta-r\cos\alpha\sin\theta=k$

$x\sin\alpha-y\cos\alpha=k$



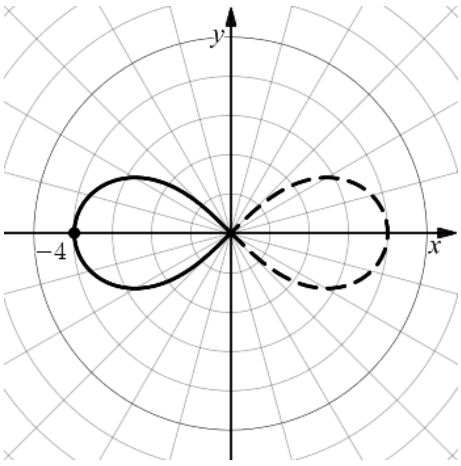
6) a)

Since $r^2 = 16\cos 2\theta$ then consider $r = \pm 4\sqrt{\cos 2\theta}$



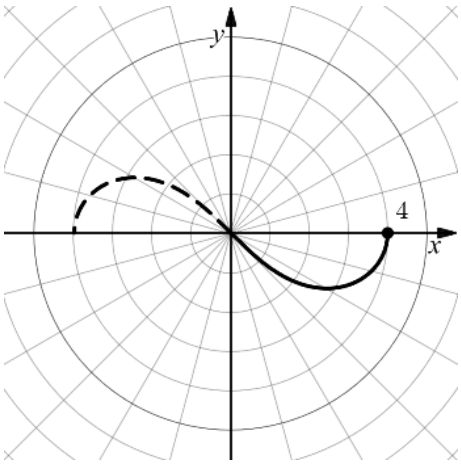
6) b)

Since $r^2 = 16\cos 2\theta$ then consider $r = \pm 4\sqrt{\cos 2\theta}$



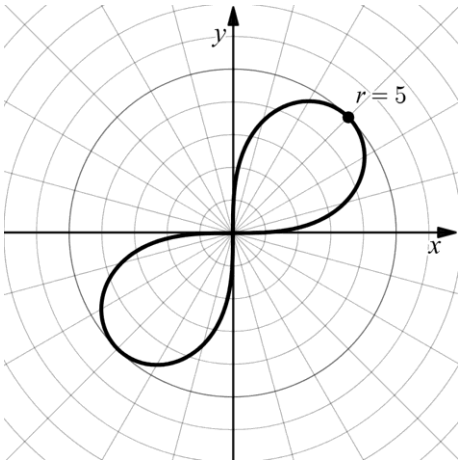
6) c)

Since $r^2 = 16\cos 2\theta$ then consider $r = \pm 4\sqrt{\cos 2\theta}$



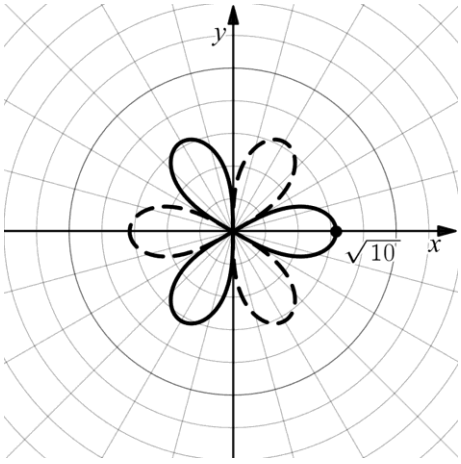
7) a)

Since $r^2 = 25\sin 2\theta$ then consider $r = \pm 5\sqrt{\sin 2\theta}$



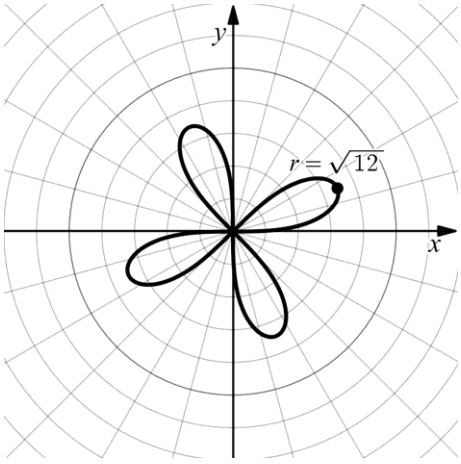
7) b)

Since $r^2 = 10\cos 3\theta$ then consider $r = \pm\sqrt{10\cos 3\theta}$



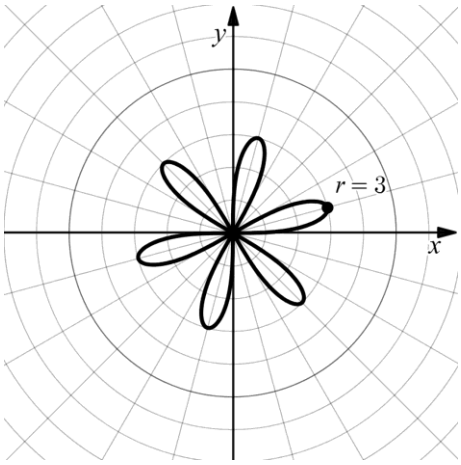
7) c)

Since $r^2 = 12\sin 4\theta$ then consider $r = \pm\sqrt{12\sin 4\theta}$



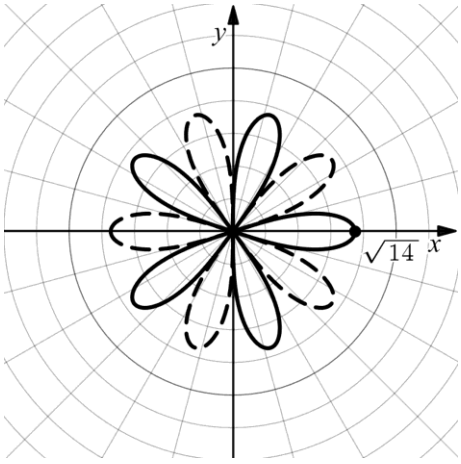
7) d)

Since $r^2 = 9\sin 6\theta$ then consider $r = \pm 3\sqrt{\sin 6\theta}$



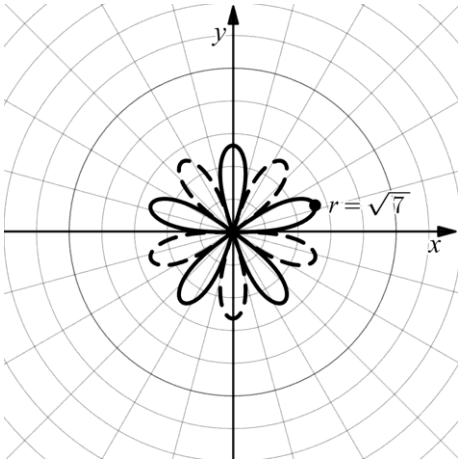
7) e)

Since $r^2 = 14\cos 5\theta$ then consider $r = \pm\sqrt{14\cos 5\theta}$



7) f)

Since $r^2 = 14\cos 5\theta$ then consider $r = \pm\sqrt{14\cos 5\theta}$



8) a)

$r^2 = 16\cos 4\theta$

8) b)

$r^2 = 9\sin 3\theta$

8) c)

$r^2 = 36\cos 5\theta$

9) a)

P occurs at the first maximum of $y = \sqrt{64\sin 3\theta} = 8\sqrt{\sin 3\theta}$

So when $\theta = \frac{\pi}{2} \div 3 = \frac{\pi}{6}$

Which gives $x = 8\cos\frac{\pi}{6}$ and $y = 8\sin\frac{\pi}{6}$

$\therefore \left(4\sqrt{3},\ 4\right)$

9) b)

Q occurs at the second maximum of $y = \sqrt{64\sin 3\theta} = 8\sqrt{\sin 3\theta}$

So when $\theta = \frac{\pi}{6} + \frac{2\pi}{3} = \frac{5\pi}{6}$

Which gives $x = 8\cos\frac{5\pi}{6}$ and $y = 8\sin\frac{5\pi}{6}$

$\therefore \left(-4\sqrt{3},\ 4\right)$

9) c)

Solve $32 = 64\sin 3\theta \Rightarrow \sin 3\theta = \frac{1}{2}$

$\theta \in \left\{\frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18}\right\}$

$r = \sqrt{32}$

$\therefore (5.57, 0.98), (3.64, 4.33), (-3.64, 4.33), (-5.57, 0.98), (-1.93, -5.32)$ and $(1.93, -5.32)$

10) a)

Consider $y = \sqrt{16\cos 4\theta} = 4\sqrt{\cos 4\theta}$

So we want values of θ for which $\cos 4\theta > 0$

So $\theta \in \left(0, \frac{\pi}{8}\right) \cap \left(\frac{3\pi}{8}, \frac{5\pi}{8}\right) \cap \left(\frac{7\pi}{8}, \pi\right)$

10) b)

Possible similarities:

They both pass through the axes at the same point.

The maximum value of $r = 4$

Both curves are symmetric about the axes.

Possible differences:

C_1 has 4 petals whilst C_2 has 8 petals.

C_2 has loops created by negative values of r

The petals of curve $C_1$are wider than the petals of C_2

C_2 is a continuous curve whilst C_1 is discontinuous.